

Harish Kumar S

Associate Software Developer

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PROFESSIONAL SUMMARY

Integrated MTech in Data Science (VIT, 2025). Associate Software Developer at FactEntry Data Solutions (A SIX Company) building production AI systems and collaborating with SIX on DME cloud migration (Databricks, data lakes, data warehouses, MLflow). Experienced in MLOps, Kubernetes, GitOps, RAG pipelines, computer vision, and scalable backend engineering. Delivered enterprise-grade AI platforms using FastAPI microservices, Redis-based distributed processing, and cloud-native deployment architectures.

Built and deployed a production bond-extraction platform on K3s, leveraging 80+ CPU cores with Redis Streams and human-in-the-loop validation workflows.

Collaborating with SIX parent company on DME migration to Databricks cloud: data lakes, data warehouses, and MLflow MLOps. Implemented end-to-end GitOps delivery pipelines using GitLab CI, Helm, and Argo CD for automated staging deployments. Developed RAG-powered document intelligence systems using FAISS, Groq LLaMA, Streamlit, and FastAPI. Published the language-processing-tool package on PyPI for reusable NLP workflows.

CORE COMPETENCIES

AI/ML Engineer Machine Learning Deep Learning Computer Vision NLP RAG Python FastAPI Kubernetes K3s Docker Redis PostgreSQL Databricks MLflow Data Lake Data Warehouse GitLab CI Argo CD Helm GitOps MLOps Roboflow YOLO FAISS LangChain Streamlit

EXPERIENCE

Associate Software Developer

FactEntry Data Solutions - A SIX Company | Dec 2025 - Present

Shipped Data Mining Engine (DME) modules on K3s stage: corporate/municipal bond extraction, validation UI, and cluster operations portal.

Collaborating with SIX parent company on DME migration to cloud: Databricks platform, data lakes, and data warehouse patterns. Supporting MLOps on Databricks with MLflow experiment/model tracking alongside existing K3s GitOps delivery. Implemented Redis-stream job dispatch with multi-node worker throughput and GitOps delivery (GitLab CI image build -> Helm chart promote -> Argo CD).

Built FastAPI microservices, live WebSocket job telemetry, role-based CPU tiering, and domain-restricted enterprise auth.

Associate Software Engineer Trainee Intern

MIMASOFT Technologies | Jun 2025 - Oct 2025

Developed Dental ML Project: Roboflow YOLOv11 tooth and panoramic X-ray models with inference CLI.

Contributed to backend integration, API workflows, and production-oriented software development practices.

MLOps and Data Science Intern

FactEntry Data Solutions - A SIX Company | Nov 2024 - May 2025

Published language-processing-tool on PyPI (PDF language detection, OCR, document clustering) as self-driven internship contribution.

Built Fixed Income Expert RAG system (FAISS, Groq LLaMA 3, Streamlit/FastAPI) as personal contribution during internship.

Data Science Intern (ML and SDE)

JP INFOTECH | Nov 2023 - Jan 2024

Built NLP web scraping with English-to-SQL query translation for extractive summarization and information retrieval.

Developed an OCR-based resume analyser with keyword suggestions and n-gram typing assist against job descriptions.

Engineered a real-time MQTT + Unity AR system to visualize environmental telemetry with ML-driven climate insights.

Applied Data Science Extern

Smart Bridge - Smartinternz | May 2023 - Jul 2023

Built a disease prediction ensemble using Random Forest and SVM for clinical trial prediction accuracy.

Stored and retrieved clinical data via MongoDB Atlas and PyMongo, transforming it into analysis-ready datasets.

EDUCATION

Integrated MTech - CSE (Data Science) | Vellore Institute of Technology | 2020 - 2025 | CGPA 7.8

Class XII | THIRU G V C Higher Secondary School | 2020 | 79%

Class X | St Joseph Matriculation Higher Secondary School | 2018 | 87.8%

TECHNICAL SKILLS

AI/ML: Machine Learning, Deep Learning, Generative AI, Computer Vision, NLP, RAG, FAISS, Sentence Transformers, Ollama, Roboflow, YOLOv11, Supervision, MLflow, Databricks

Engineering: Python, FastAPI, Streamlit, SQL, Docker, Kubernetes (K3s), Redis, PostgreSQL, Databricks, Data Lake, Data Warehouse, LangChain, PyMuPDF, OpenCV

DevOps/GitOps: GitHub, GitLab CI, Argo CD, Helm, systemd, GitHub Actions, GitOps, MLflow

KEY PROJECTS

Data Mining Engine (DME)

FactEntry Data Solutions (A SIX Company) - Associate Software Developer | K3s, FastAPI, Redis, PostgreSQL, Argo CD, Databricks, MLflow
Enterprise bond extraction on K3s with GitOps CI/CD; collaborating with SIX on Databricks cloud migration (data lakes, warehouses, MLflow).

Recruiting Agent - AI-Powered ATS

Independent R&D - Agents Monorepo | FastAPI, React, SQLite, ChromaDB, Ollama, Groq

Multi-step LLM ATS - resume OCR, ChromaDB JD matching, verification, suspicion scoring, and HR feedback loop.

Trip Guide - Conversational Trip Planner

Independent R&D - Agents Monorepo | FastAPI, Google ADK, Gemini, Groq, Ollama, Leaflet

Multi-agent travel assistant - ADK orchestrator with weather/places/stays specialists, SSE chat, live trip board + OSRM map.

Fixed Income Expert System

FactEntry internship - personal contribution | FastAPI, Streamlit, FAISS, Groq, SBERT

RAG chatbot for financial bond PDFs - upload, embed, retrieve, LLaMA 3 answers via Groq.

Cybersecurity Content Moderator

VIT - Capstone project (academic) | LLM, Streamlit, Docker, FastAPI

Multimodal hate-speech and harmful content classification (text, image, PDF).

Language Distribution & Document Clustering Tool

FactEntry internship - self contribution | Python, Tesseract, LangDetect, PyPI

PyPI-published PDF language detector, OCR, and clustering pipeline (v0.2.7).

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals

Applied Data Science (SmartInternz + Google Developers)

Data Analysis and Visualization with Power BI

Preparing Data for Analysis with Microsoft Excel

Harnessing the Power of Data with Power BI

Extract, Transform, and Load Data in Power BI